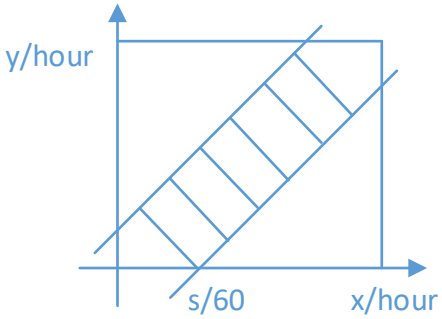
Solutions to PSP Exam 2026

B—bookwork, E—new example, T—new theory

1. a)



See the square in the figure, the origin is 5:00pm, x axis (unit: hour) represents the time Alice arrive and y axis represents the time Bob arrives. The length of the square is 40/60 = 2/3. [3E]

So, "each is prepared to wait s/60 hours before leaving and they meet each other" can be mathematically described as

$\left\{|x-y| \le \frac{s}{60}\right\}$ (see the shadow area in the figure) [3E]

Thus

$$P\left(|x-y| \le \frac{s}{60}\right) = \iint_{|x-y|\le\frac{s}{60}} f(x,y)\,dxdy = 1 - \frac{9}{4} \times \left(\frac{2}{3} - \frac{s}{60}\right)^2 \quad [3E]$$

where(x, y) is a two-dimensional variable stands for the time Alice and Bob arrive, and its density function is

$$f(x,y) = \begin{cases} \frac{9}{4}, & 0 \le x \le \frac{2}{3},\ 0 \le y \le \frac{2}{3} \\ 0, & \text{otherwise} \end{cases} \quad [3E]$$

Finally,

$$1 - \frac{9}{4} \times \left(\frac{2}{3} - \frac{s}{60}\right)^2 \ge \frac{3}{4} \Rightarrow s \ge 20 \text{ min} \quad [3E]$$

b)

From the independence assumption,

[2T]

$$F(y) = P(Y \leq y) = P(X_1 \leq y) \cdot \ldots \cdot P(X_n < y)$$

$$= \frac{1}{2}y \cdot \ldots \cdot \frac{1}{2}y \quad \text{[2T]}$$

$$= \frac{1}{2^n}y^n \quad \text{[2T]}$$

Thus,

$$f(y) = \frac{n}{2^n}y^{n-1}, \quad \text{[2T]}$$

$$y \in [0,2] \quad \text{[2T]}$$

2.

a) The joint distribution of n observations (where $\bar{x} = 6$ is the empirical mean)

[2E]

$$f(X,\beta) = \frac{\beta^{n\alpha}(x_1 \dots x_n)^{(\alpha-1)}e^{-n\beta\bar{x}}}{\Gamma^{\mathrm{n}}(\alpha)}, \quad x > 0, \alpha = 10, \beta > 0$$ [2E]

Take partial derivative:

$$\frac{\partial f(X,\beta)}{\partial \beta} = \frac{n\alpha\beta^{n\alpha-1}(x_1 \dots x_n)^{(\alpha-1)}e^{-n\beta\bar{x}}}{\Gamma^{\mathrm{n}}(\alpha)} - n\bar{x}\frac{\beta^{n\alpha}(x_1 \dots x_n)^{(\alpha-1)}e^{-n\beta\bar{x}}}{\Gamma^{\mathrm{n}}(\alpha)}$$ [2E]

$$= (\frac{n\alpha}{\beta} - n\bar{x})f(X,\beta)$$

[2E]

Thus, letting it be 0, we have

$$\beta = \frac{\alpha}{\bar{x}} = \frac{10}{6} = \frac{5}{3}$$ [2E]

b) Recall the Wiener-Hopf equation

$$\mathbf{c} = \mathbf{R}^{-1}\mathbf{r}$$

$$\sigma^2 = r_0 - \mathbf{r}^T\mathbf{R}^{-1}\mathbf{r}$$

i) If n=1, the Wiener-Hopf equation trivially reads

[2E]

$$R_Y(0)c_1 = R_Y(1)$$

Therefore,

$$c_1 = \frac{R_Y(1)}{R_Y(0)} = \frac{J_0(2\pi f_d)}{J_0(0)} = J_0(2\pi f_d)$$

[2E]

because $J_0(0) = 1$. Therefore,

$$c_1 = J_0(2\pi \times 0.3) = 0.291$$

$$\sigma^2 = 1 - r_1^2 = 1 - 0.291^2 = 0.915$$

[2E]

ii) When n = 2, we have

$$\mathbf{R} = \begin{bmatrix} 1 & 0.291 \\ 0.291 & 1 \end{bmatrix}$$

$$\mathbf{r} = \begin{bmatrix} -0.402 \\ 0.291 \end{bmatrix} \quad \text{[3E]}$$

Thus the coefficient vector

[3E]

$$\mathbf{c} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \mathbf{R}^{-1}\mathbf{r} = \begin{bmatrix} 1 & 0.291 \\ 0.291 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -0.402 \\ 0.291 \end{bmatrix} = \begin{bmatrix} -0.532 \\ 0.446 \end{bmatrix}$$

Mean-square error

$$\sigma^2 = 1 - [-0.402 \quad 0.291] \begin{bmatrix} 1 & 0.291 \\ 0.291 & 1 \end{bmatrix}^{-1} \begin{bmatrix} -0.402 \\ 0.291 \end{bmatrix} = 0.657 \quad \text{[3E]}$$

3.

a) i) It's clear that $E[X(n)] = 0$ and $\mathrm{Var}[X(n)] = 1$. The autocorrelation function [2E]

$$R_X(m, m+n) = \cos(n\lambda)\cos\big((n+m)\lambda\big) + \sin(n\lambda)\sin\big((n+m)\lambda\big) = \cos(m\lambda) \quad [2E]$$

ii) Clearly it is wide-sense stationary. [2E]

iii) In general, this random process is not strict-sense stationary. To see this, let's consider an example where $\lambda = \frac{\pi}{2}$. Then

[2E]

$$X(n) = A\cos(n\frac{\pi}{2}) + B\sin(n\frac{\pi}{2})$$

Obviously, the samples at different times

$$X(1) = B$$
$$X(2) = -A \quad [2E]$$

have different distributions.

The answer to part (iii) isn't unique. Any reasonable justification will get credit.

b)

i) This is the same as the average time when the third student arrives.

$$E[T_4] = \frac{3}{\lambda} = \frac{3}{0.2} = 15\ minutes \qquad \text{[3E]}$$

ii) This means that the number of students arriving in the first half an hour is less than 3.

Let t = 15 minutes. Then $\lambda t = 0.2 \times 15 = 3$ and $e^{-\lambda t} = e^{-3}$. Recall N(t) has Poisson distribution:

$$P(N(t) = k) = e^{-\lambda t}\frac{(\lambda t)^k}{k!}, \qquad k = 0,1,2,\ldots \qquad \text{[3E]}$$

Therefore,

$$P(N(t) < 3) = P(N(t) = 0) + P(N(t) = 1) + P(N(t) = 2) \qquad \text{[3E]}$$

$$= e^{-3} + 3e^{-3} + 4.5e^{-3} = 0.423$$

iii) Let t_1 = 5 minutes, t_2 = 10 minutes. Since a Poisson process is memoryless,

$$P(N(t_1) \geq 1)P(N(t_2) - N(t_1) \leq 2) \qquad \text{[2E]}$$

$$= P(N(t_1) \geq 1)P(N(t_2 - t_1) \leq 2)$$

$$= [1 - P(N(t_1) = 0)][P(N(t_2 - t_1) = 0) + P(N(t_2 - t_1) = 1) + P(N(t_2 - t_1) = 2) \qquad \text{[2E]}$$

$$= [1 - e^{-1}]\left[e^{-1} + e^{-1} + \frac{1}{2}e^{-1}\right] = 0.58 \qquad \text{[2E]}$$

Note here that $\lambda t_1 = \lambda(t_2 - t_1) = 1$ and $e^{-\lambda t_1} = e^{-\lambda(t_2 - t_1)} = e^{-0.2\times 5} = e^{-1}$.

4.

a) From the equation $\pi = \pi P$, we obtain

$\pi_1 = \frac{2}{3}\pi_1 + \frac{1}{3}\pi_2 \qquad \Rightarrow \pi_1 = \pi_2$ [2E]

$\pi_2 = \frac{1}{3}\pi_1 + \frac{1}{3}\pi_2 + \frac{1}{3}\pi_3 \qquad \Rightarrow \pi_2 = \pi_3$ [2E]

…

$\pi_M = \frac{1}{3}\pi_{M-1} + \frac{2}{3}\pi_M \qquad \Rightarrow \pi_{M-1} = \pi_M$ [2E]

So all the π_i's are equal, and

$$\pi_i = \frac{1}{M} \text{ for all } i.$$ [2E]

b)

i) A martingale satisfies the following property

$$E[X_{n+1}|X_0, \dots, X_n] = X_n$$ [1E]

which we need to verify.

Since $\{X_n\}$ is a Markov chain, we have

$$E[X_{n+1}|X_0, \dots, X_n] = E[X_{n+1}|X_n]$$

Now, using [2E]

$$p_{jk} = \binom{N}{k}\left(\frac{j}{N}\right)^k\left(1 - \frac{j}{N}\right)^{N-k}$$

We obtain

$$E[X_{n+1}|X_n] = \sum_{k=0}^{N} k p_{X_n k}$$

$$= \sum_{k=0}^{N} k\binom{N}{k}\left(\frac{X_n}{N}\right)^k\left(1 - \frac{X_n}{N}\right)^{N-k}$$ [3T]

Since p_{jk} is a binomial distribution, the above mean is given by

$$N\frac{X_n}{N} = X_n$$ [2T]

Therefore,

$$E[X_{n+1}|X_0, \dots, X_n] = E[X_{n+1}|X_n] = X_n$$

which implies $\{X_n\}$ forms a martingale.

ii)

Again by the Markov property, we have

$$E[Y_{n+1}|X_0, \dots, X_n] = E[Y_{n+1}|X_n] \quad \text{[2E]}$$

We obtain

$$E[Y_{n+1}|X_n] = \sum_{k=0}^{N} k(N-k)p_{X_n k}$$

$$= \sum_{k=0}^{N} k(N-k)\binom{N}{k}\left(\frac{X_n}{N}\right)^k\left(1-\frac{X_n}{N}\right)^{N-k} \quad \text{[2T]}$$

We break this up into two terms:

$$N\sum_{k=0}^{N} k\binom{N}{k}\left(\frac{X_n}{N}\right)^k\left(1-\frac{X_n}{N}\right)^{N-k} - \sum_{k=0}^{N} k^2\binom{N}{k}\left(\frac{X_n}{N}\right)^k\left(1-\frac{X_n}{N}\right)^{N-k} \quad \text{[3T]}$$

The first term is N times the mean:

$$= NX_n$$

The second term is the second moment of a binomial distribution:

$$= X_n + \left(1-\frac{1}{N}\right)X_n^2$$

Therefore,

$$E[Y_{n+1}|X_n] = NX_n - X_n - \left(1-\frac{1}{N}\right)X_n^2 \quad \text{[2T]}$$

$$= Y_n\left(1-\frac{1}{N}\right)$$

which $\neq Y_n$, implying $\{Y_n\}$ does not form a martingale.

However, if we define

$$S_n = Y_n/\left(1-\frac{1}{N}\right)^n$$

It will form a martingale.